



NAT

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NAT

NAT to technologia sieciowa, która pozwala na ukrycie całej sieci lokalnej (używającej prywatnych adresów IP) za jednym lub kilkoma publicznymi adresami IP. Serwer z włączonym NAT działa jak pośrednik: zamienia adresy źródłowe pakietów wychodzących z sieci lokalnej na swój własny adres zewnętrzny, dzięki czemu stacje robocze mogą korzystać z Internetu.

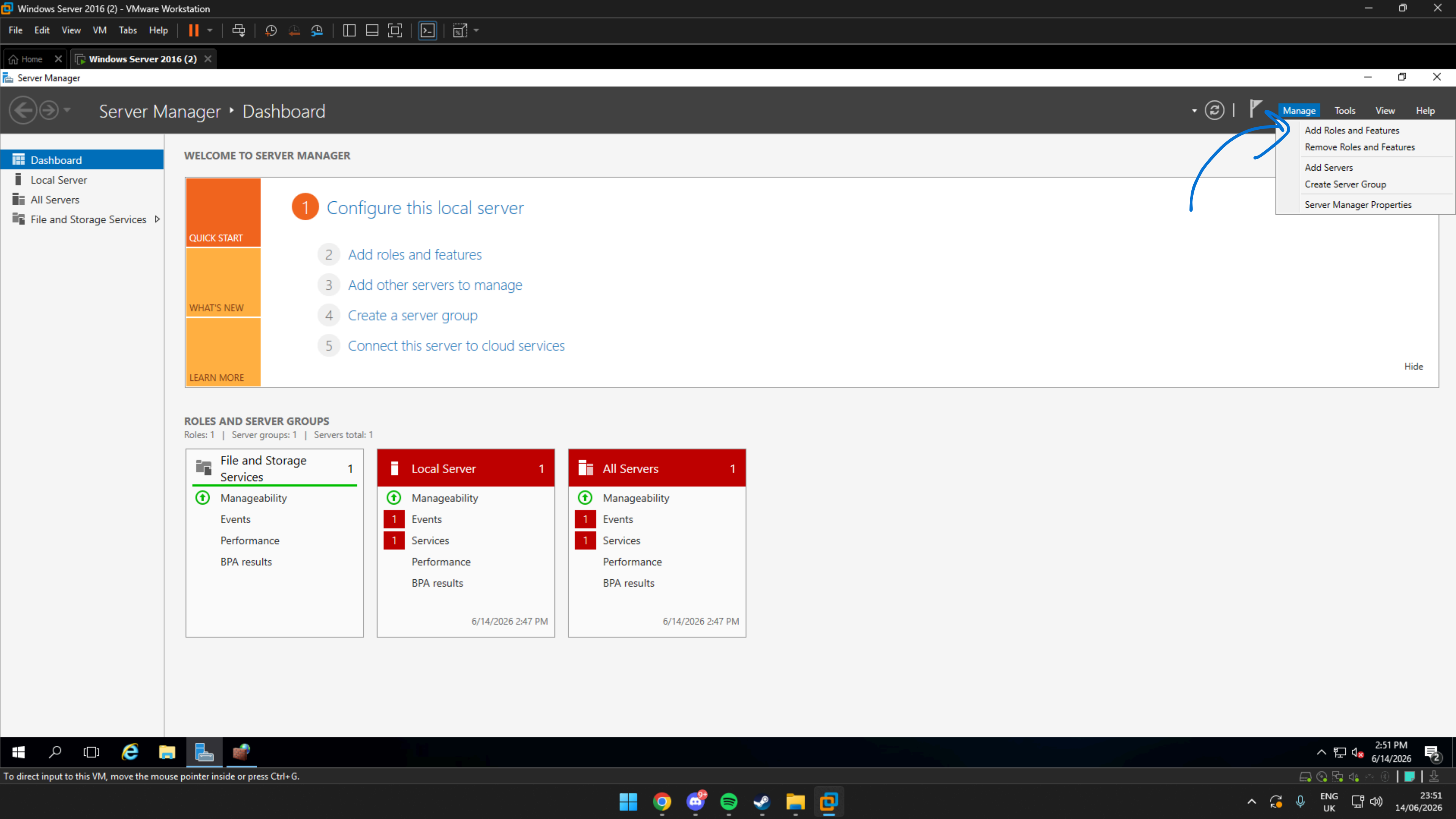
DWIE KARTY SIECIOWE

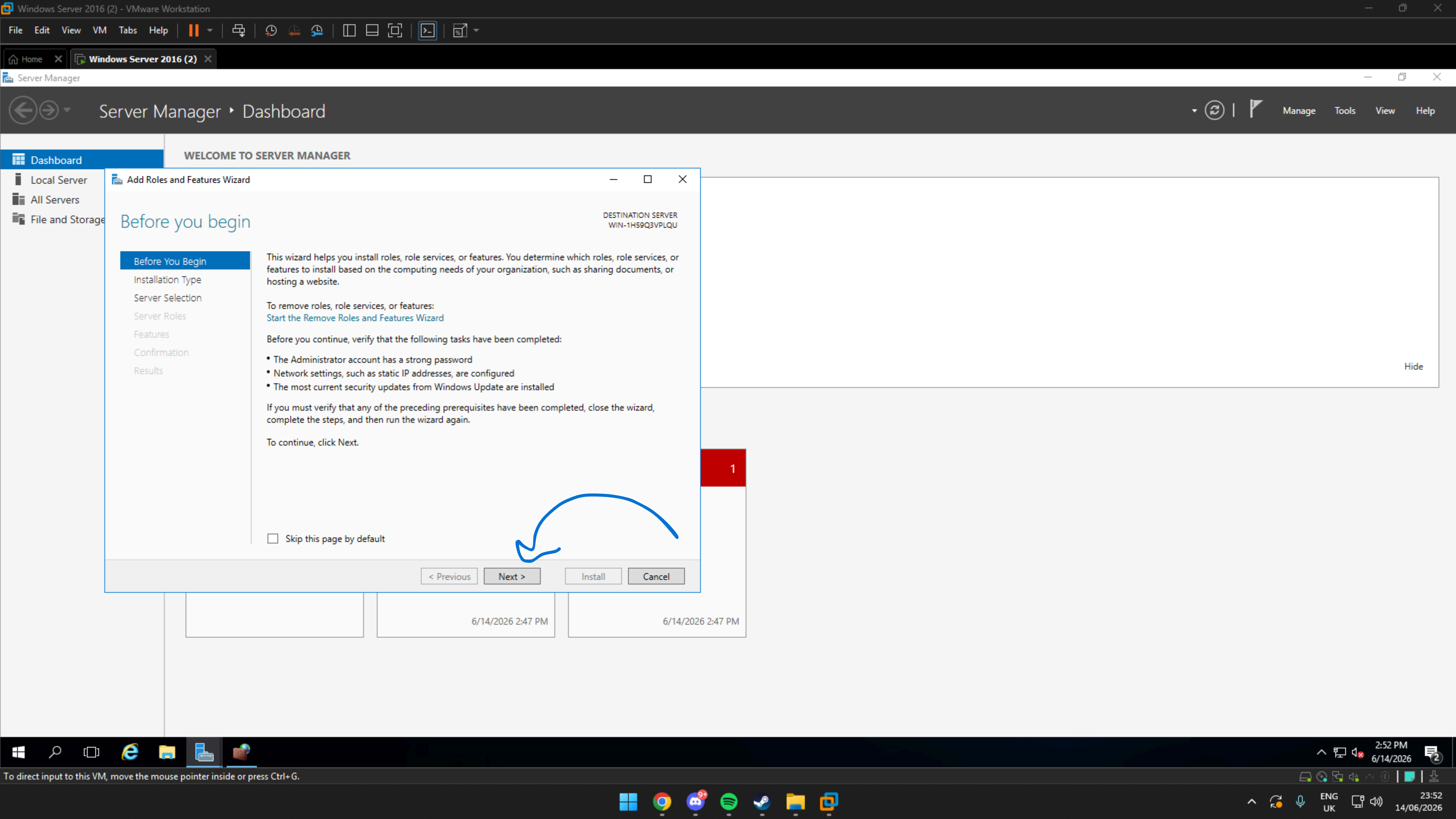
Żeby w ogóle móc skonfigurować NAT, Twój Windows Server w VirtualBoxie musi mieć dodane dwie karty sieciowe:

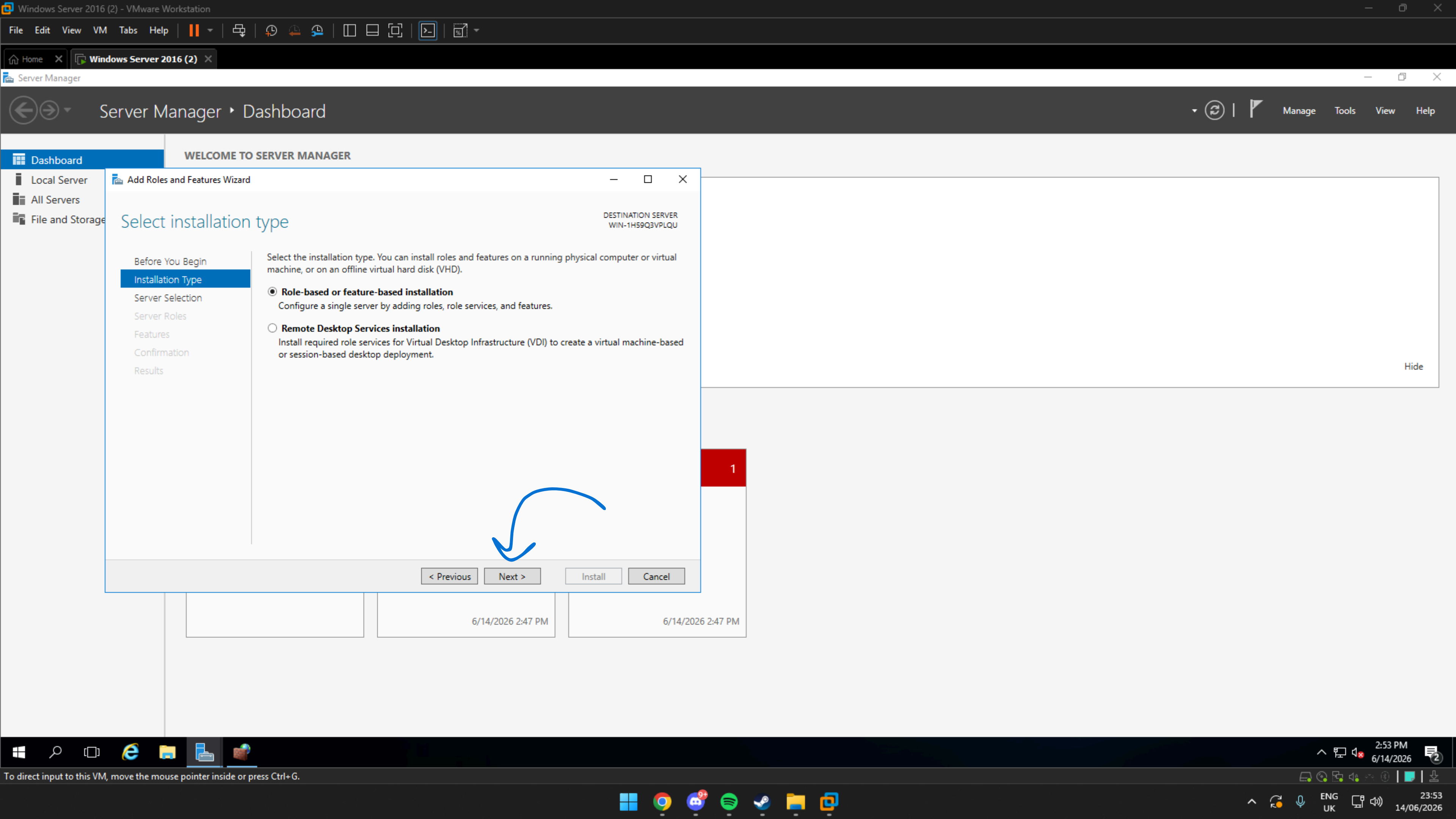
- Karta wewnętrzna (LAN): Skonfigurowana w sieci wewnętrznej (Internal Network), np. z adresem 192.168.89.2. Do niej podpięty jest Twój klient.
- Karta zewnętrzna (WAN): Skonfigurowana zazwyczaj jako Mostkowana (Bridged) lub NAT w ustawieniach VirtualBoxa. To ona ma dostęp do prawdziwego Internetu (pobiera adres z Twojego domowego routera).

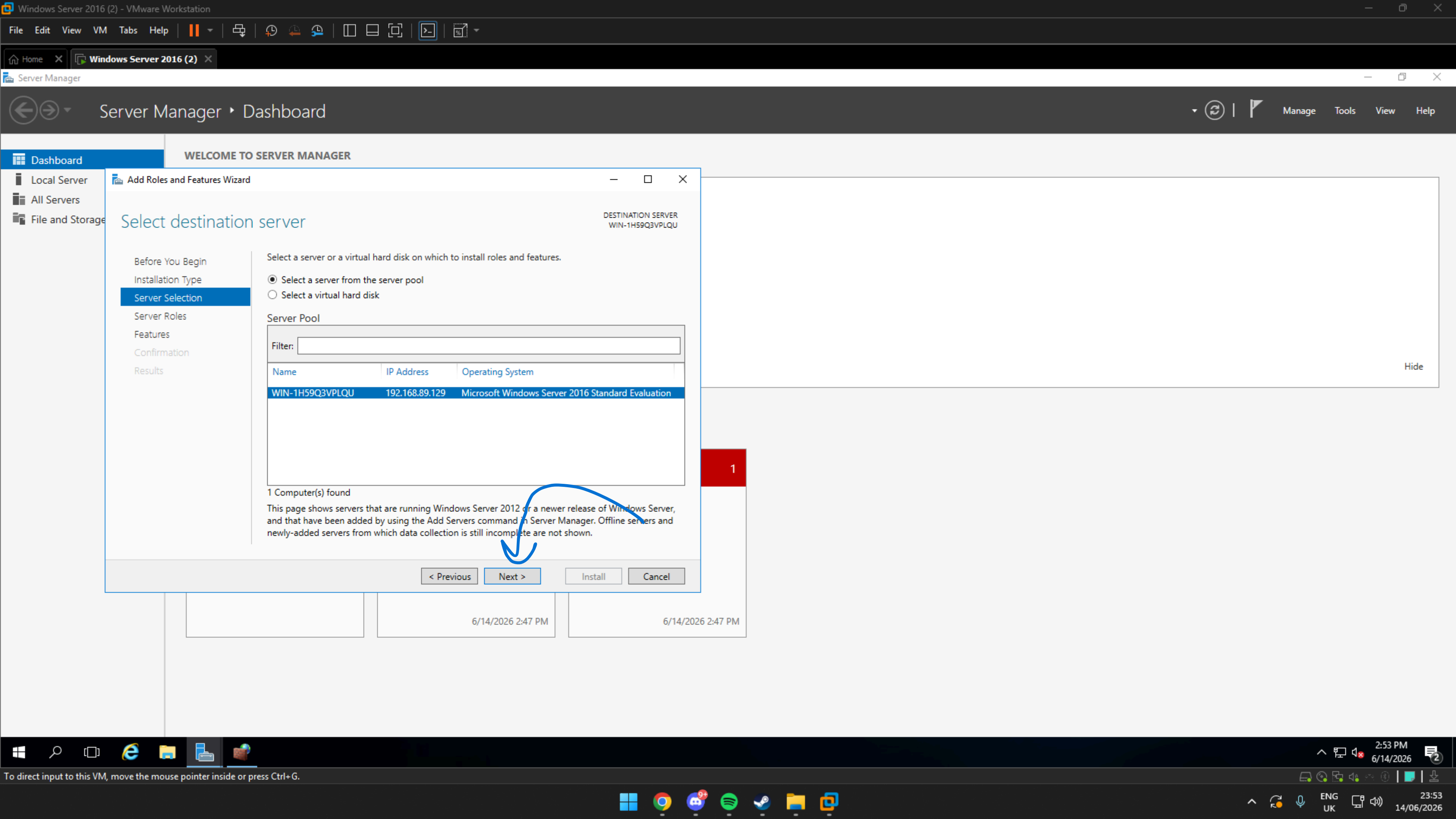
KONFIGURACJA NAT

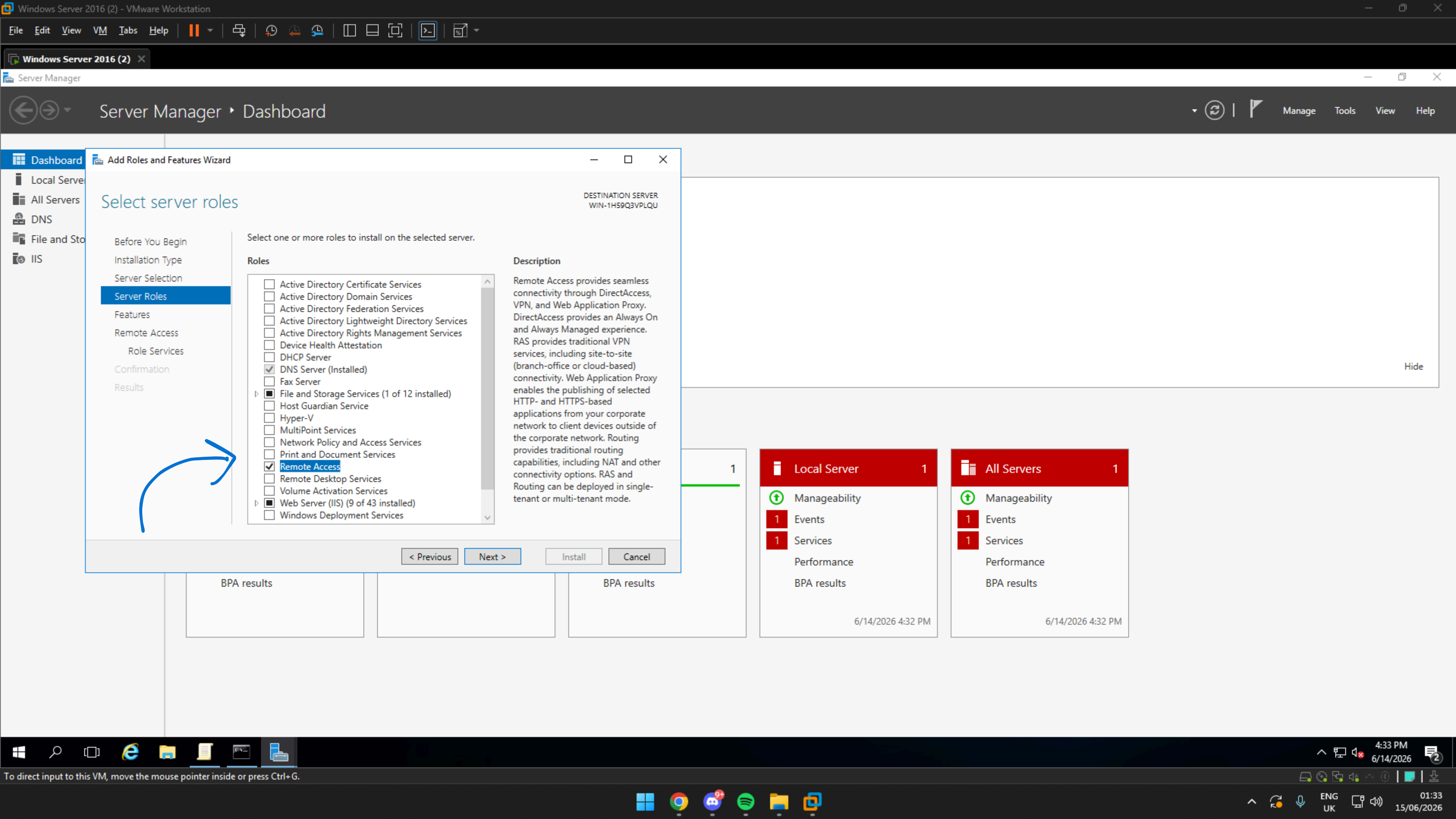
TREŚĆ ZADANIA: ZAINSTALUJ I SKONFIGURUJ USŁUGĘ ROUTING I DOSTĘP ZDALNY. URUCHOM PROTOKÓŁ NAT NA KARCIE SIECIOWEJ PODŁĄCZONEJ DO INTERNETU (KARTA ZEWNĘTRZNA), ABY UMOŻLIWIĆ STACJOM ROBOCZYM DOSTĘP DO SIECI ZEWNĘTRZNEJ.

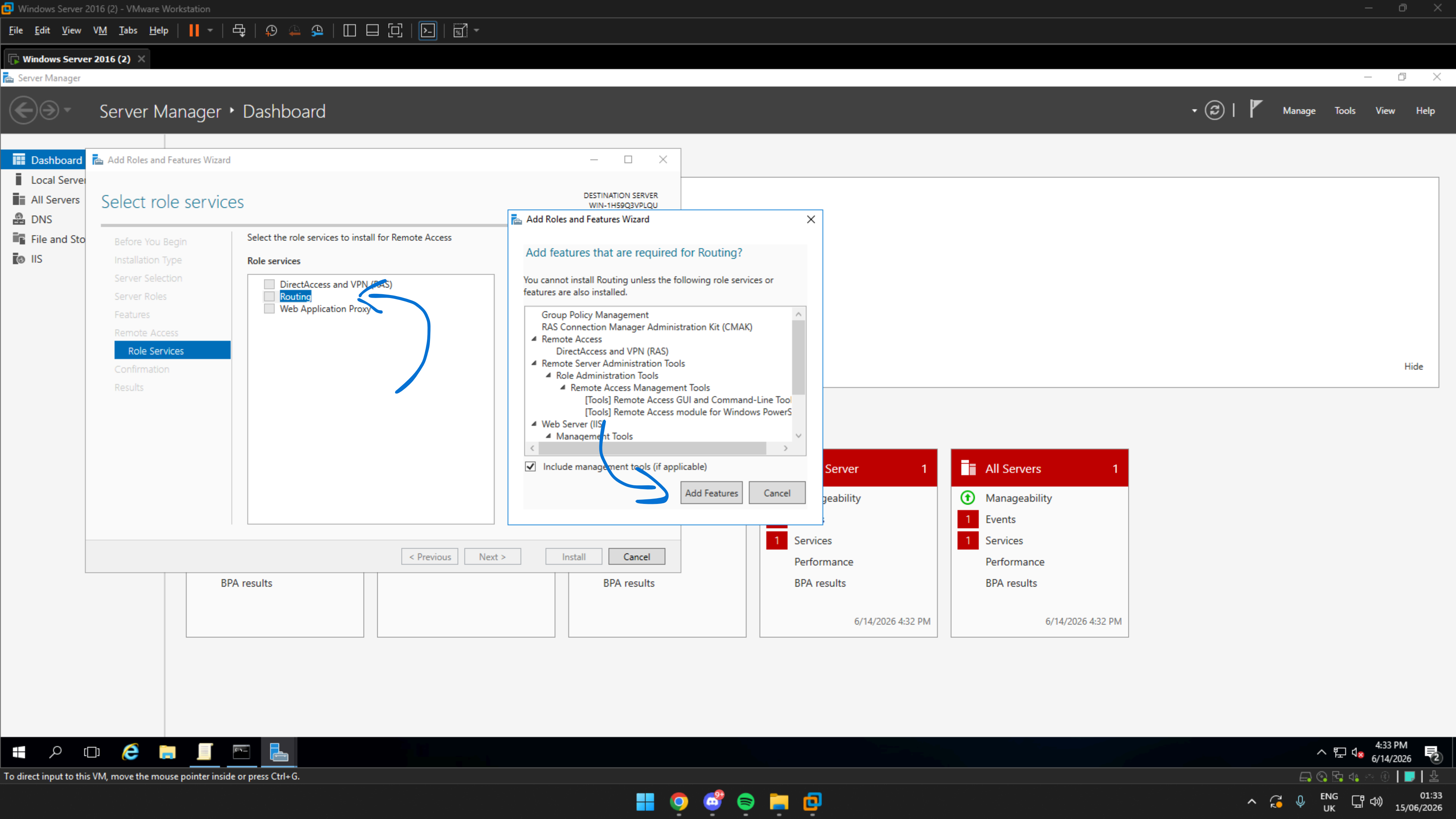


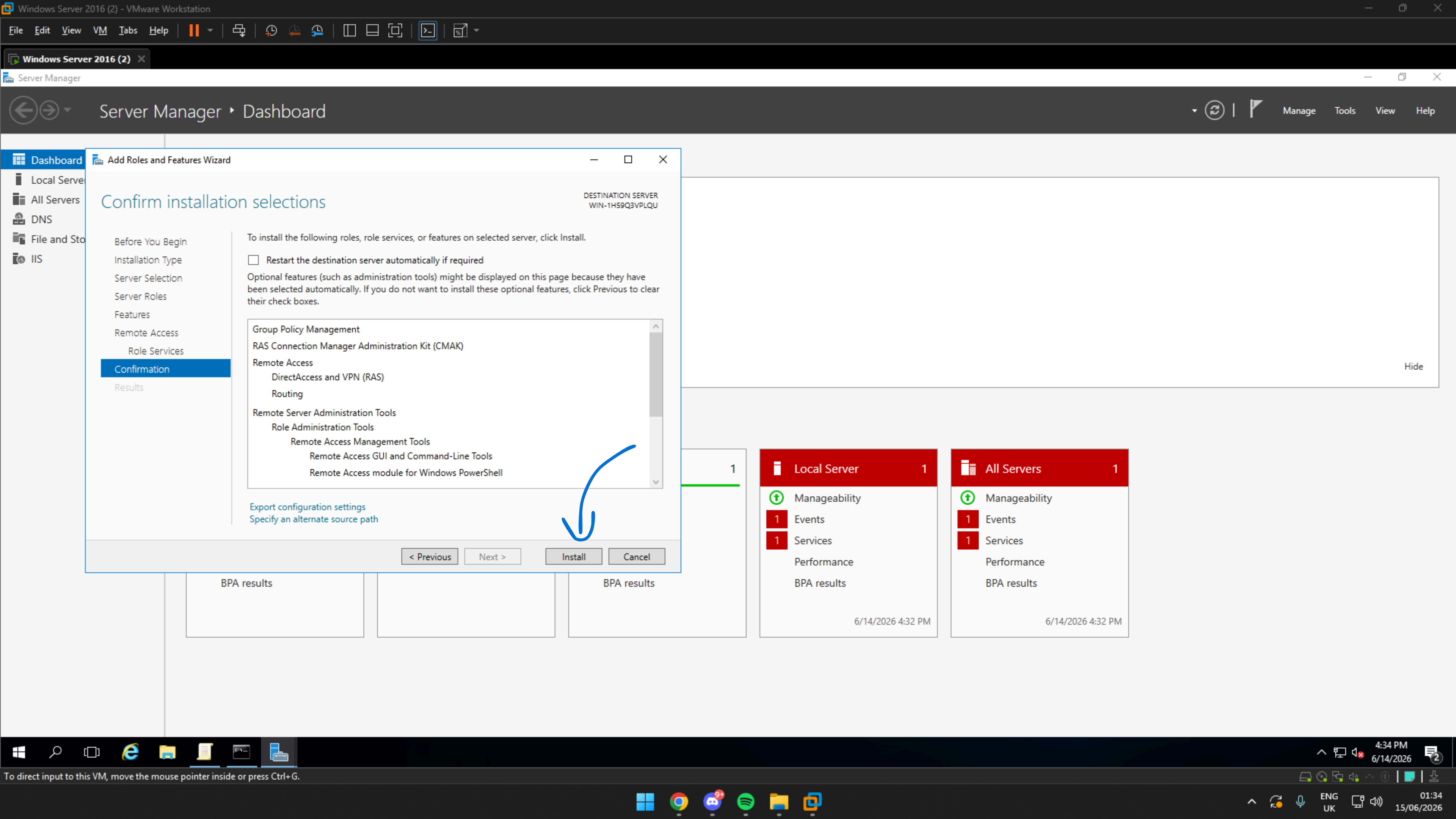


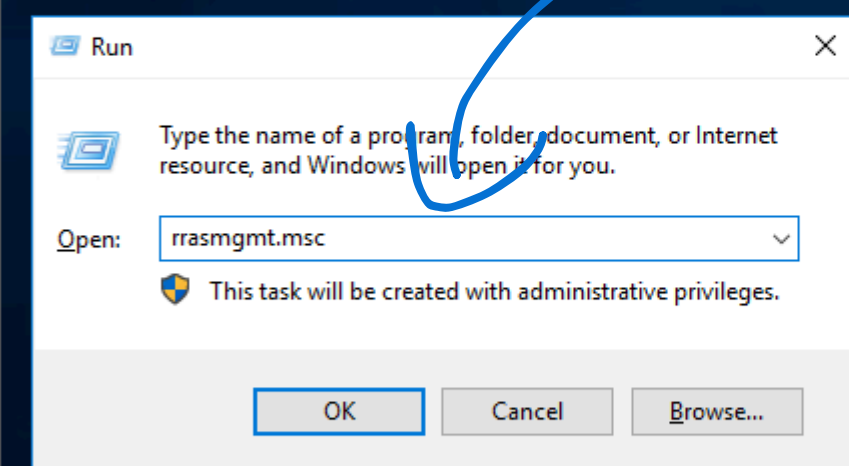


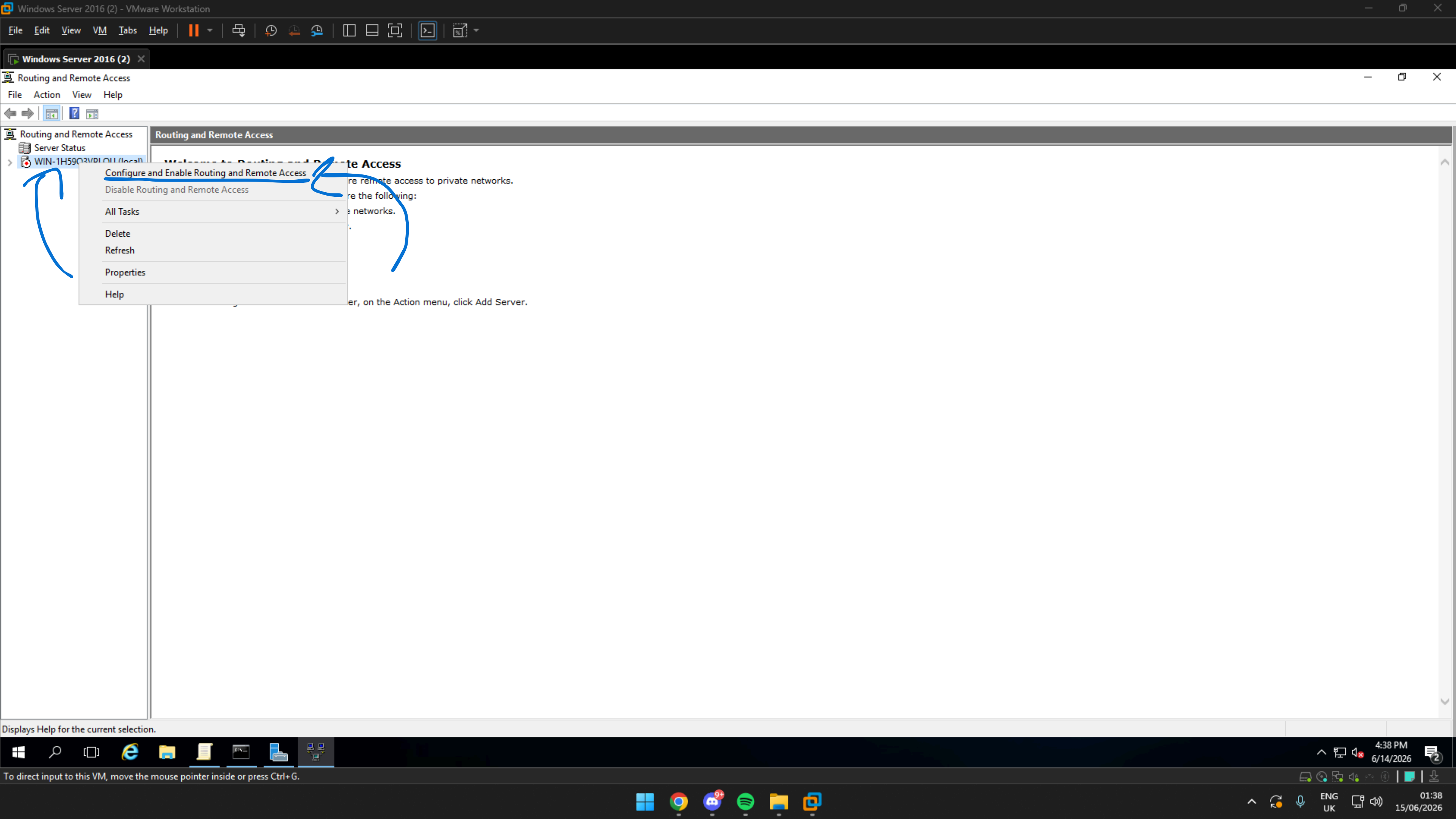


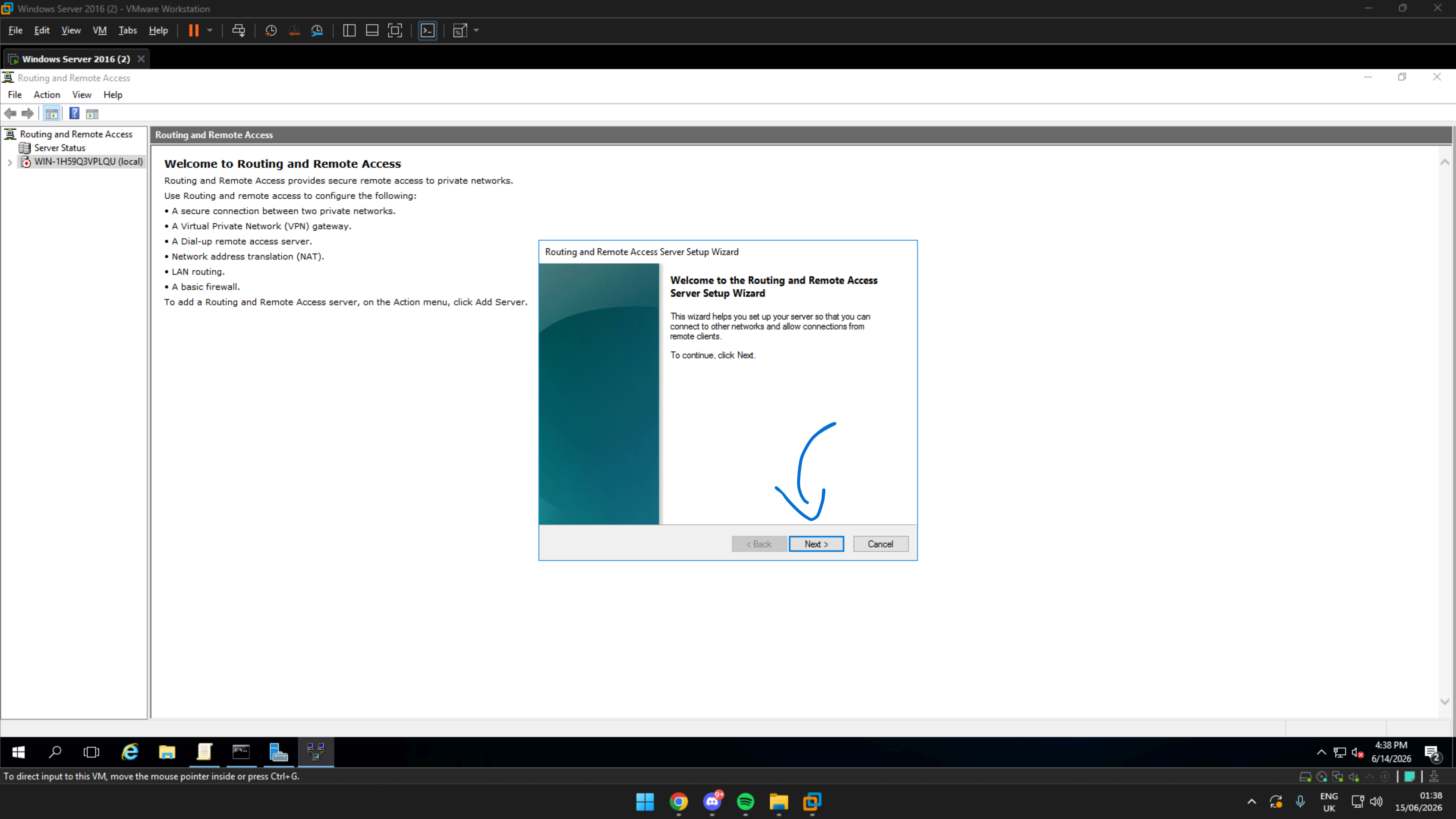


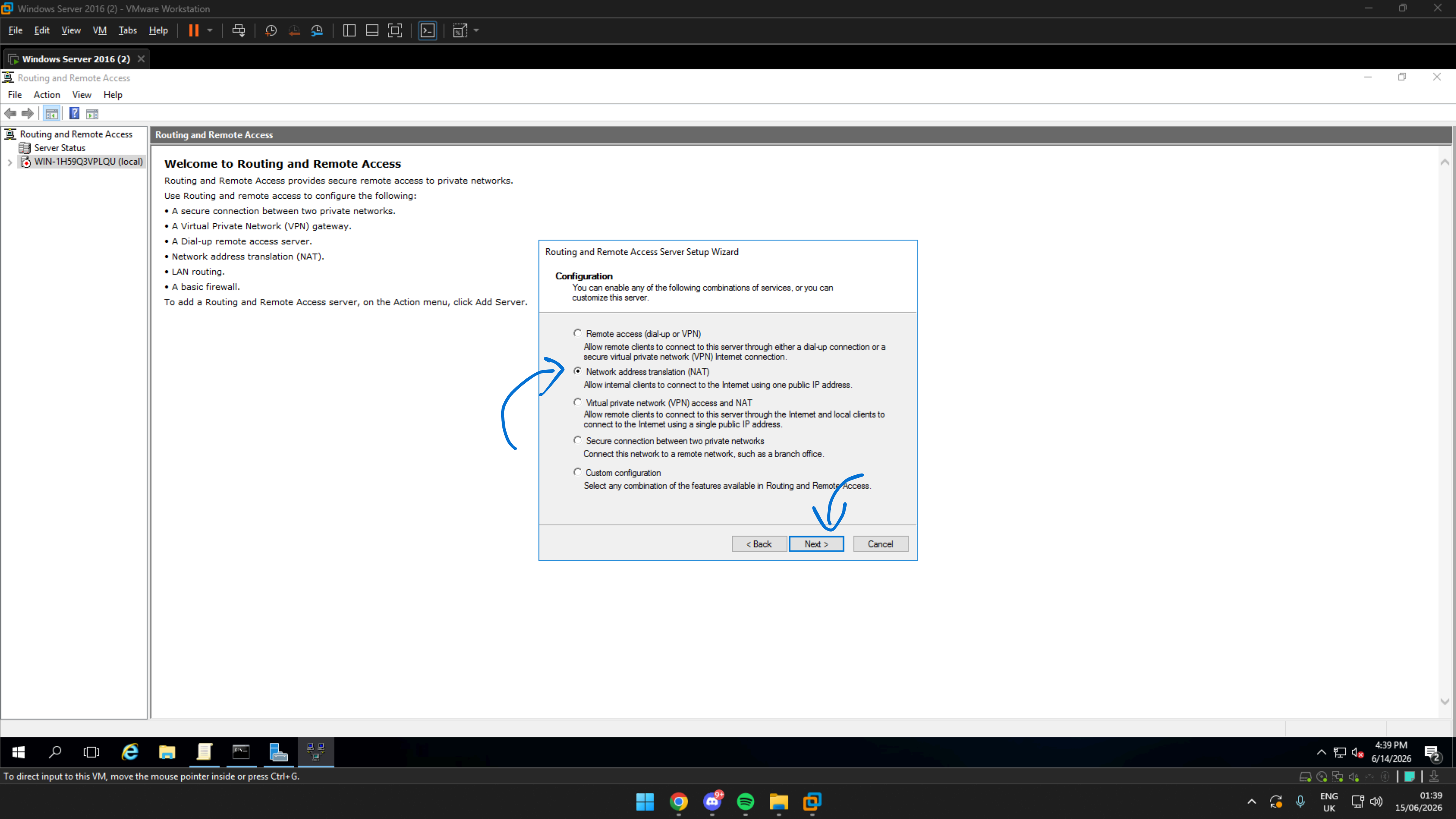












Routing and Remote Access

Welcome to Routing and Remote Access

Routing and Remote Access provides secure remote access to private networks.

Use Routing and remote access to configure the following:

- A secure connection between two private networks.
- A Virtual Private Network (VPN) gateway.
- A Dial-up remote access server.
- Network address translation (NAT).
- LAN routing.
- A basic firewall.

To add a Routing and Remote Access server, on the Action menu, click Add Server.

Routing and Remote Access Server Setup Wizard

NAT Internet Connection

You can select an existing interface or create a new demand-dial interface for client computers to connect to the Internet.

- ☒ Use this public interface to connect to the Internet:

Network Interfaces:

Name	Description	IP Address
Ethernet0	Intel(R) 82574L Gigabit...	192.168.89.129 (DHCP)
Ethernet1	Intel(R) 82574L Gigabit...	192.168.0.1

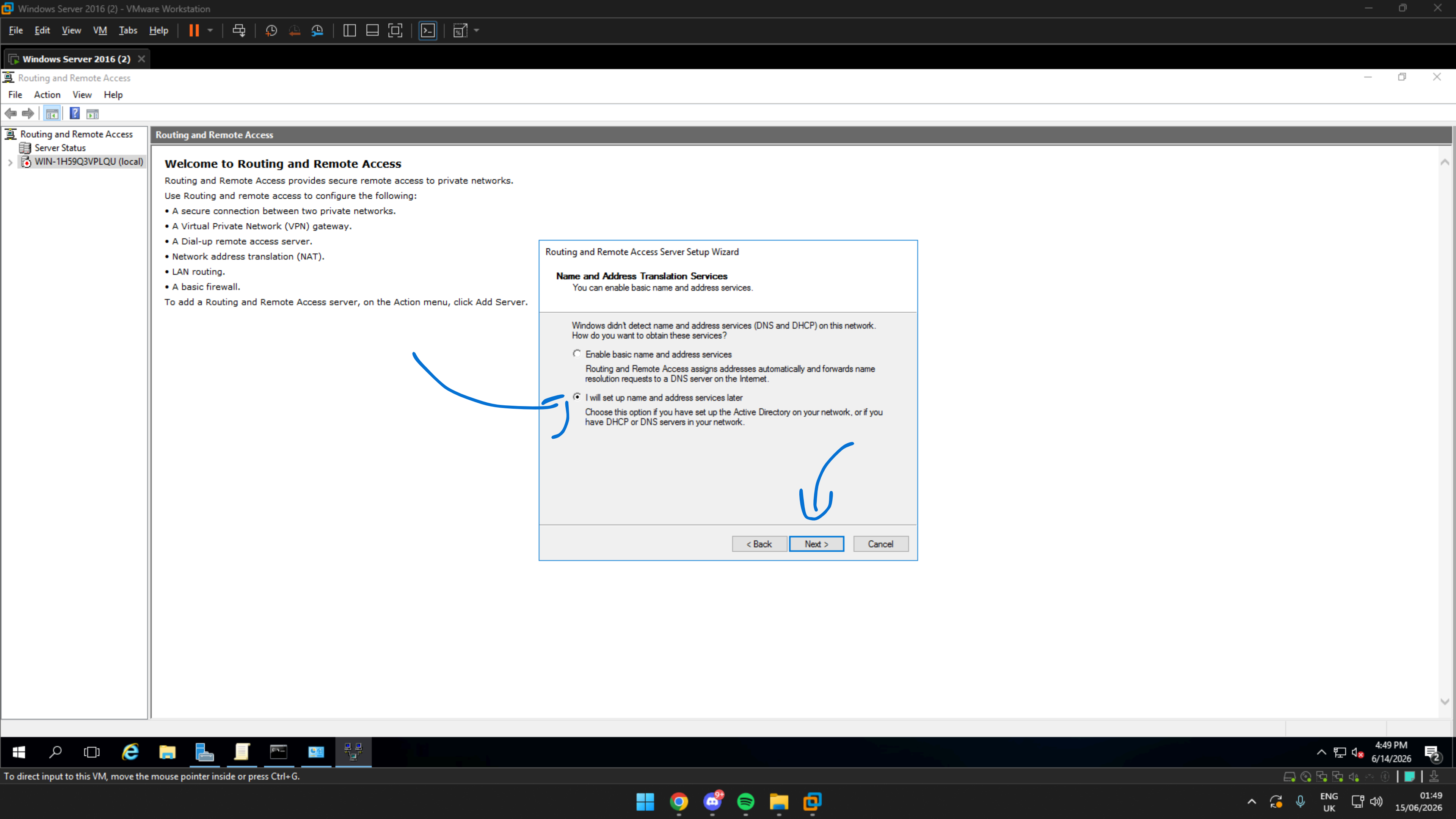
- ☐ Create a new demand-dial interface to the Internet

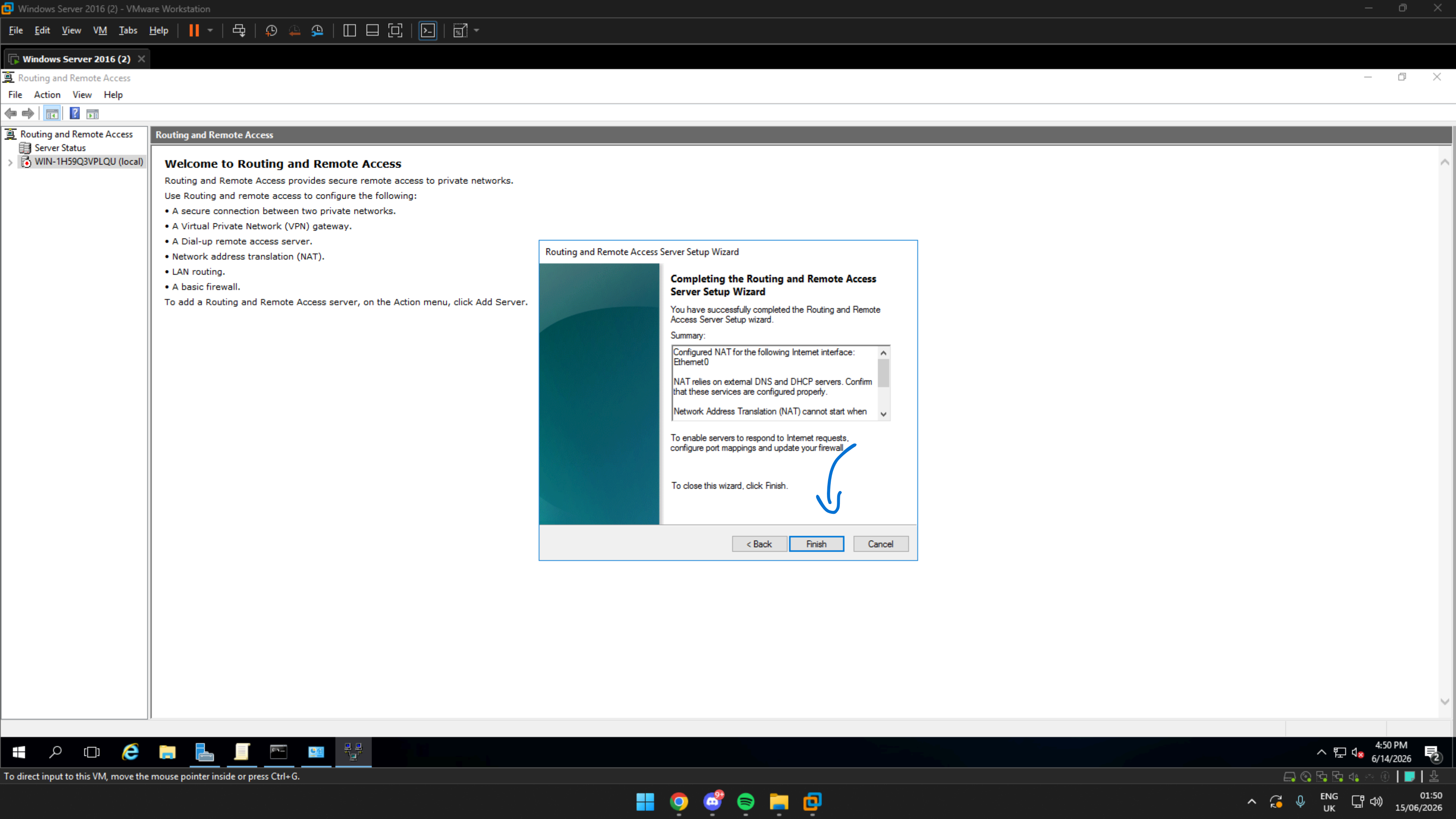
A demand-dial interface is activated when a client uses the Internet. Select this option if this server connects with a modem or by using the Point-to-Point Protocol over Ethernet. The Demand-Dial Interface Wizard will start at the end of this wizard.

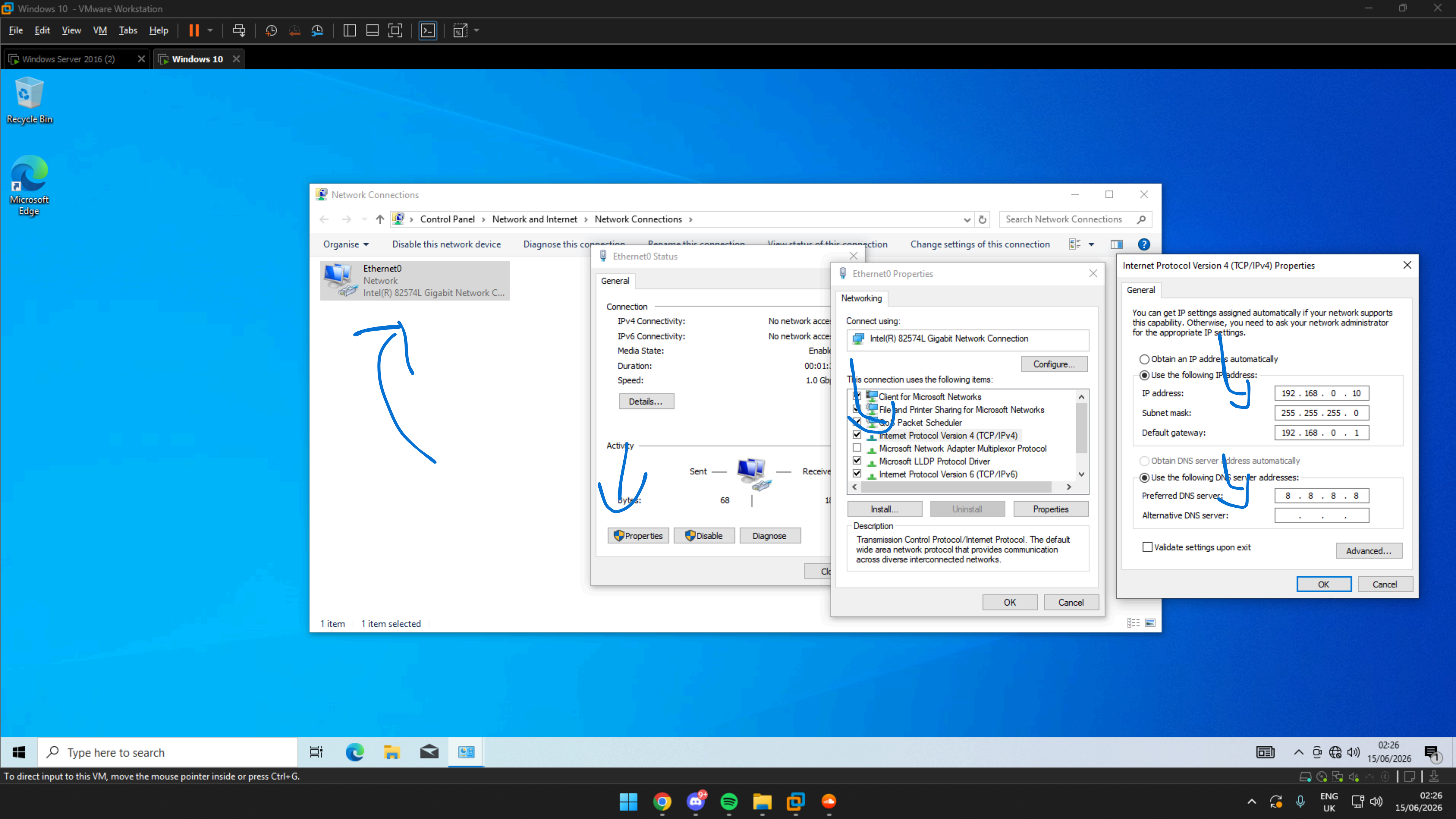
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Cancel







Windows Server 2016 (2) - VMware Workstation

File Edit View VM Tabs Help

Windows Server 2016 (2)

Windows Firewall with Advanced Security

File Action View Help

Inbound Rules

Outbound Rules

Connection Security Rules

Monitoring

Name	Group	Profile
Core Networking - IPv6 (IPv6-In)	Core Networking	All
Core Networking - Multicast Listener Do...	Core Networking	All
Core Networking - Multicast Listener Qu...	Core Networking	All
Core Networking - Multicast Listener Rep...	Core Networking	All
Core Networking - Multicast Listener Rep...	Core Networking	All
Core Networking - Neighbor Discovery A...	Core Networking	All
Core Networking - Neighbor Discovery S...	Core Networking	All
Core Networking - Packet Too Big (ICMP...	Core Networking	All
Core Networking - Parameter Problem (I...	Core Networking	All
Core Networking - Router Advertisement...	Core Networking	All
Core Networking - Router Solicitation (IC...	Core Networking	All
Core Networking - Teredo (UDP-In)	Core Networking	All
Core Networking - Time Exceeded (ICMP...	Core Networking	All
Cortana	Cortana	All
DHCPv4 Relay Agent [Client] (UDP-In)	DHCP Relay Agent	All
DHCPv6 Relay Agent [Server] (UDP-In)	DHCPv6 Relay Agent	All
DIAL protocol server (HTTP-In)	DIAL protocol server	Private
DIAL protocol server (HTTP-In)	DIAL protocol server	Domain
Distributed Transaction Coordinator (RPC)	Distributed Transaction Co...	All
Distributed Transaction Coordinator (RP...	Distributed Transaction Co...	All
Distributed Transaction Coordinator (TCP...	Distributed Transaction Co...	All
DNS (TCP, Incoming)	DNS Service	All
DNS (UDP, Incoming)	DNS Service	All
RPC (TCP, Incoming)	DNS Service	All
RPC Endpoint Mapper (TCP, Incoming)	DNS Service	All
File and Printer Sharing (Echo Request - I...	File and Printer Sharing	All
File and Printer Sharing (Echo Request - I...	File and Printer Sharing	All
File and Printer Sharing (LLMNR-UDP-In)	File and Printer Sharing	All
File and Printer Sharing (NB-Datagram-In)	File and Printer Sharing	All

Inbound Rules

New Rule...

Filter by Profile

Filter by State

Filter by Group

View

Refresh

Export List...

Help

Selected Items

Disable Rule

Cut

Copy

Delete

Help

02:32 15/06/2026

Windows 10 - VMware Workstation

File Edit View VM Tabs Help

Windows 10

Recycle Bin

Microsoft Edge

Network Connections

Ethernet0
Network
Intel(R) 82574L

1 item | 1 item selected

Command Prompt

C:\Users\admin>ping 192.168.0.1

Pinging 192.168.0.1 with 32 bytes of data:

Reply from 192.168.0.1: bytes=32 time<1ms TTL=128

Reply from 192.168.0.1: bytes=32 time<1ms TTL=128

Reply from 192.168.0.1: bytes=32 time<1ms TTL=128

Reply from 192.168.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\admin>

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